RAK15007 WisBlock 1MByte FRAM Module Datasheet

Overview

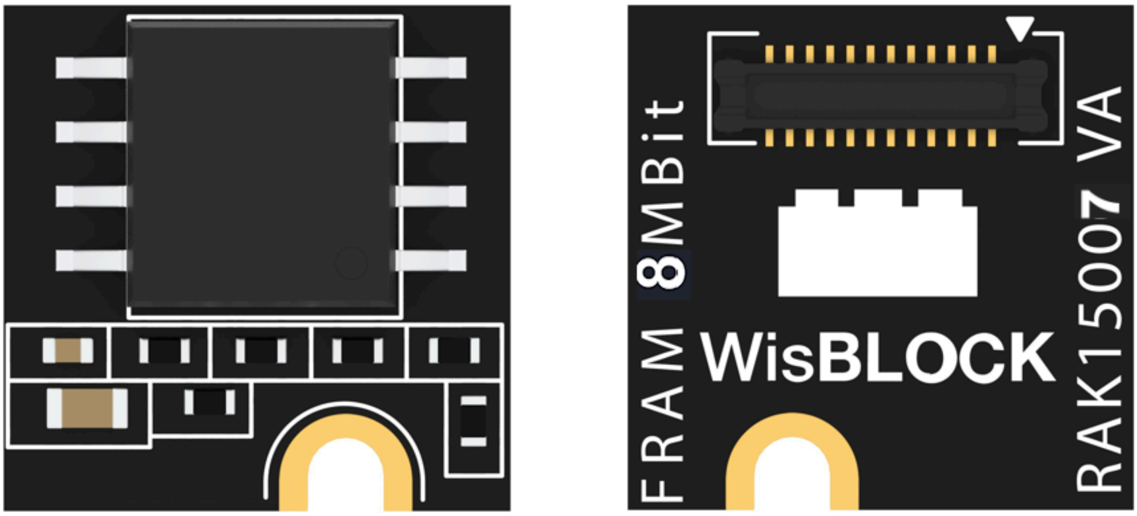

Figure 1: RAK15007 WisBlock 512kByte FRAM Module

Description

RAK15007 is a WisBlock 1MByte FRAM module that extends the WisBlock system with an CY15B108QN memory module from Cypress. This module is interfaced via SPI. Additionally, it can be mounted to the sensor slot of the WisBlock Base board.

Product Features

- Sensor specifications
  - Temperature Range：-40 °C to +85 °C
  - SPI compatible digital interface, supports 40 MHz

- Operating power supply current: 2.6 mA (Typical @ 40 MHz)
- Standby current: 3.5 uA (Max)
- Sleep current: 0.9 uA (Max)
- Logically organized as 1,048,576 × 8 bits
- High Reliability：
  - Read/write endurance： 1,000,000,000,000,000/byte
  - Data retention：
    - 10 years(+85 °C)
    - 141 years(+70 °C)
    - 151 years(+60 °C)
    - 160 years(+50 °C)

- **Size**
  - 10 × 10 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK15007 module on a WisBlock Base board. The RAK15007 module can be mounted on the slots: **A, C, D, E & F**.

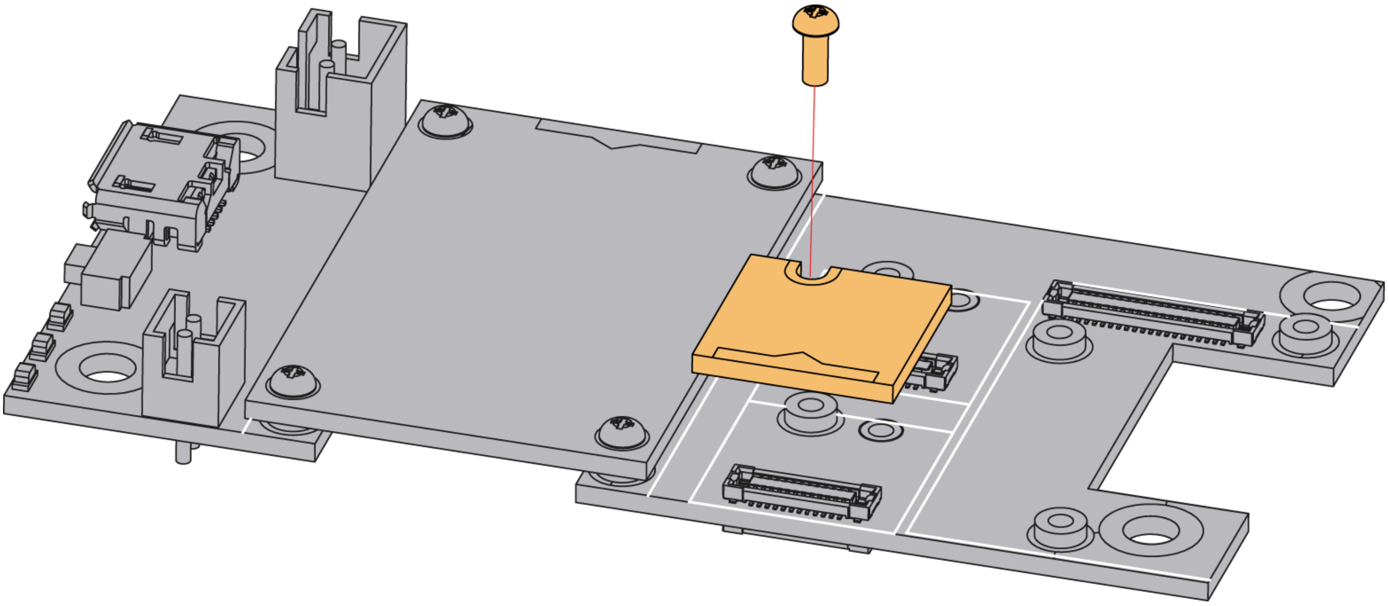

Figure 2: RAK15007 WisBlock FRAM Module Mounting

Hardware

The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK15007 WisBlock FRAM Module.

Chipset

Vendor	Part Number
Cypress	CY15B108QN

Pin Definition

The RAK15007 WisBlock 1MByte FRAM Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK15007 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

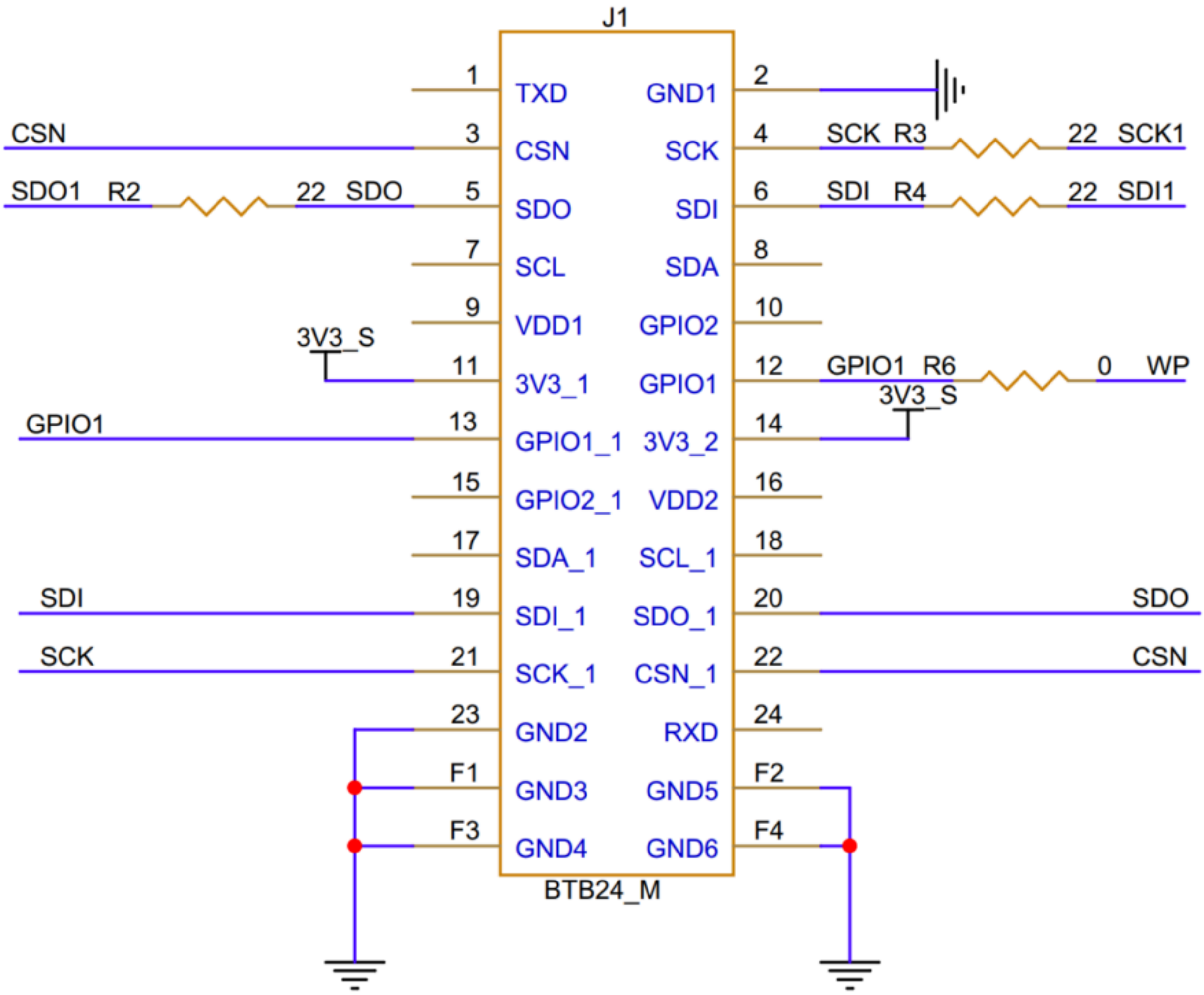

Figure 3: RAK15007 WisBlock FRAM Module Pinout Diagram

The **WisBlock Sensor** connector is used to this module and the IO used for **WP** pin at **Pin 12** will depend on where sensor slot the module is plugged in. The table shows the compatible pins used by different sensor slots:

WP (Write Protect Pin)

Slot A	Slot C	Slot D	Slot E	Slot F
IO1	IO3	IO5	IO4	IO6

> **NOTE**

> **RAK15007** should not be plugged in **Slot B** as it is dedicated for **IO2 (WisBlock IO2 pin)** that controls the **3V3_S** voltage output.

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
VDD	Power supply for the module		3.3		V
Istb	Standby current	-	-	3.5	uA
Izz	Sleep current	-	-	0.9	uA
Idd	Operating power supply current	-	-	2.6	mA

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK15007 module.

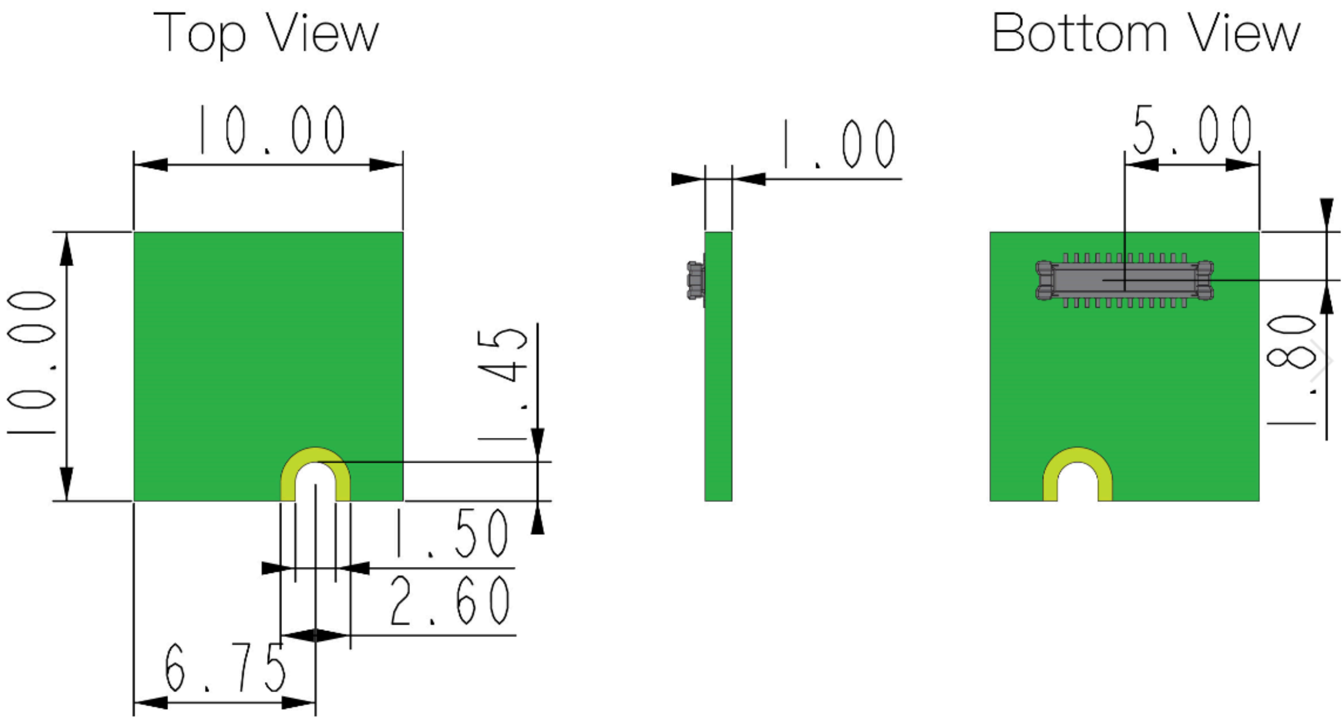

Figure 4: RAK15007 WisBlock FRAM Module Mechanic Drawing

WisConnector PCB Layout

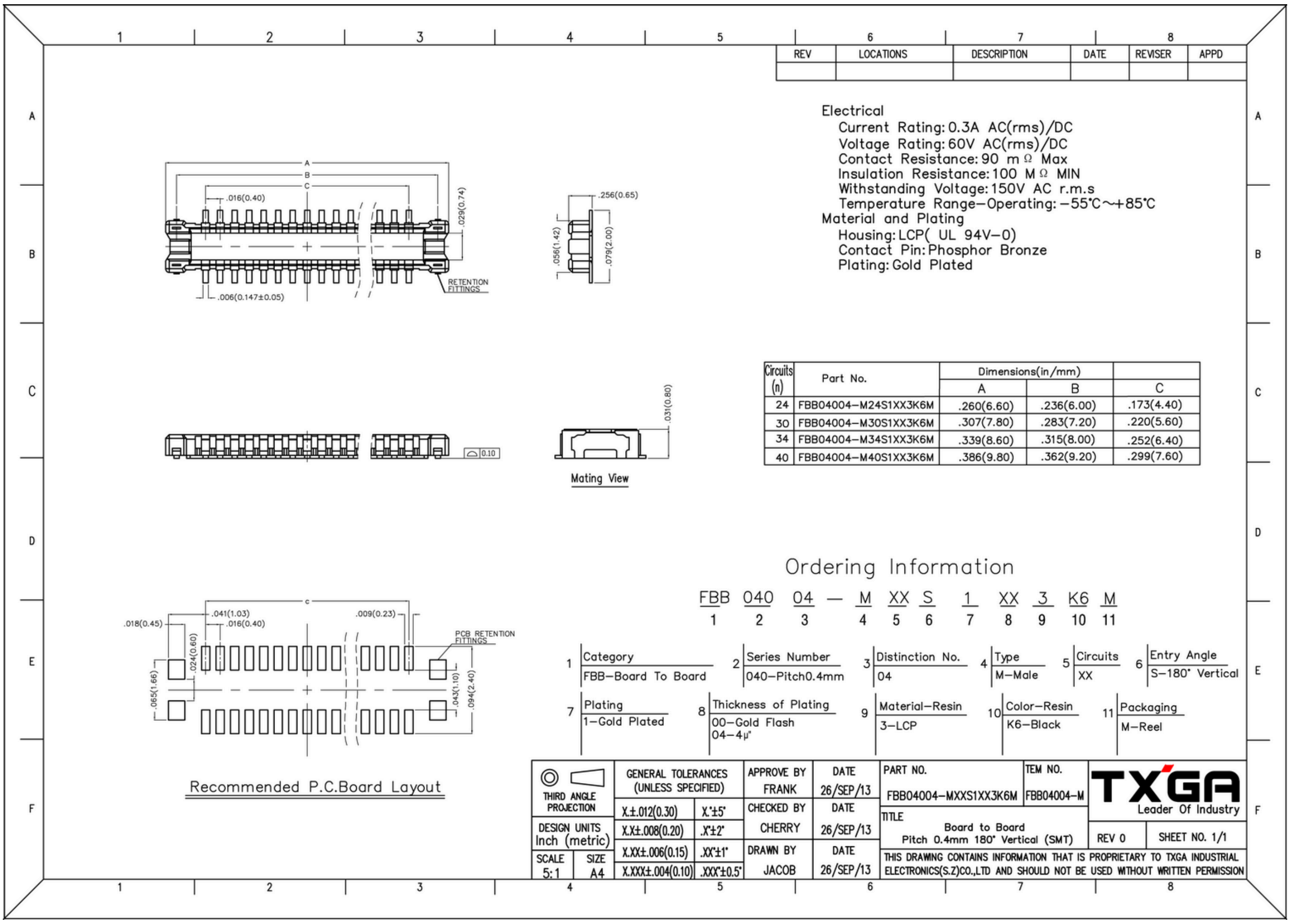

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK15007 module.

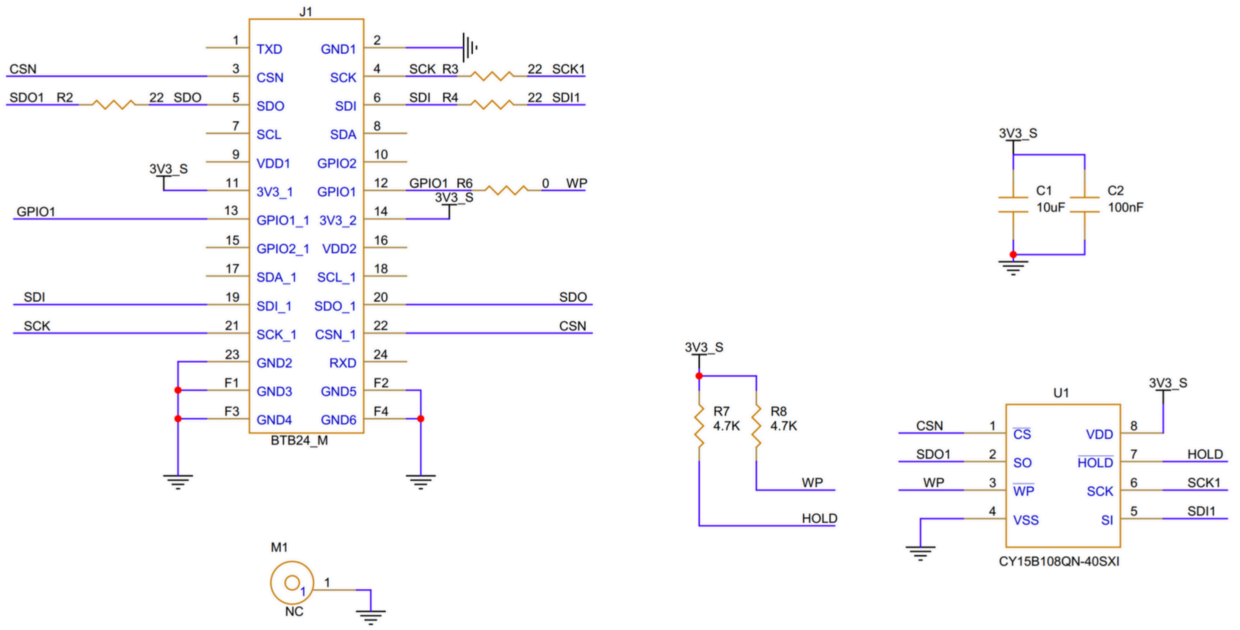

Figure 6: RAK15007 WisBlock FRAM Module schematics